

GUAVION

Executive Summary

Guavion is building the governed execution system for AI era workflows. It is the platform that **turns user intent into reliable, auditable, and reproducible work**. Instead of forcing teams to piece together software, scripts, infrastructure, and oversee disconnected systems, Guavion unifies the entire execution stack in a single environment, providing a workspace for creating and managing workflows; an AI layer that understands goals and orchestrates tasks; a containerized software stack for reproducible execution; dynamic code and script generation **tailored to individual job(s)**; scalable, optimized cloud and HPC infrastructure; and an audit layer that records, improves, and personalizes execution over time by learning from the user. We **begin in science**, where complexity, compute intensity, and reproducibility are critical, and expand into other workflow-heavy industries facing the same underlying infrastructure gap.

THE PROBLEM

Across science, finance, and enterprise environments, teams spend, on average, **30 to 60% of their time on workflow overhead** rather than on the work itself. That time is lost to fragmented tools, brittle scripts, manual configuration, and processes that are difficult to reproduce or audit. AI copilots can help generate answers, steps, or code, but generation alone does not solve execution. In serious technical environments, workflows must also be validated, run correctly, governed in real time, and made fully traceable afterward. That end-to-end execution infrastructure does not exist in a unified form today.

THE SOLUTION

Guavion solves this by unifying **the full path from intent to execution**, in the following steps:

- At the foundation is the workspace layer, which acts as the operational environment for technical work. This is where workflows are created, reviewed, edited, monitored, and managed.
- On top of that sits the AI layer. This is the reasoning engine that interprets user intent, breaks work into actionable steps, recommends methods, and governs execution as tasks unfold.
- Guavion then connects that intelligence to containerized software environments (permissive and non-permissive software code), **making the right tools portable, reproducible, and easy to deploy** without the usual burden of installation, dependency management, and fragile configuration.
- It also generates the code and scripts required to carry out the work. Instead of trapping users inside static templates, Guavion creates the execution logic needed for the specific problem, workflow, and context.
- Once the method is ready, Guavion orchestrates execution across optimized cloud and HPC environments, **selecting the optimal compute resources** for the scale, cost, and performance requirements of **the individual job**.
- Finally, every workflow is captured in an audit and learning layer that records what was requested, what decisions were made, what tools were used, what code was generated, what data flowed through the system, and what outcomes were produced. This makes execution reproducible, defensible, and continuously improvable for each user and organization.

The result is a closed-loop system that does not just assist with work, but also structures it, executes it, governs it, and learns from it (**user expertise and intelligence are absorbed and deposited in the user workspace over time**).

WHY GUAVION WINS

- **vs. Big Tech (Microsoft, AWS, Google):** they build general-purpose horizontal APIs. They do not own the validated workflow layer: the domain-specific templates, compliance guardrails, and institutional memory that enterprises actually need. Depth beats breadth here.
- **vs. Vertical Tools (Schrödinger, Benchling):** they are domain-locked. Guavion is the orchestration layer above them, and can orchestrate their outputs.
- **vs. RPA & Open Source (UiPath, Nextflow):** rule-based and static. Guavion adapts **dynamically** and governs execution in ways open-source pipelines cannot.

Guavion's advantage is that it owns the full path from intent to governed execution. Every workflow run **strengthens** the platform through better templates, richer **provenance**, stronger validation logic, and user-specific learning. In regulated industries, governed and reproducible execution is not a nice-to-have. It is a structural requirement. That makes Guavion's position **durable** even as general-purpose AI becomes more capable.

MARKET & BUSINESS MODEL

TAM	\$100B+ Workflow automation, RPA, enterprise AI, cloud orchestration
SAM	~\$12.5B 50,000 workflow-heavy enterprises × \$250K avg. annual spend
Wedge	~\$4B+ Chemoinformatics (\$3.7B) + ELN (\$659M→\$966M)
SOM (5yr)	~\$37.5M ARR 250 customers × \$150K ACV

Revenue: Enterprise SaaS (\$100K–\$250K ACV) + compute orchestration margin + vertical workflow templates. Science is the right wedge: regulated, compute-intensive workflows demand the highest level of governance, proving here earns trust to expand into finance, simulation, and enterprise operations.

TRACTION

- **Production-scale prototype:** AI-driven multi-step execution validated on molecular systems up to ~500K atoms across HPC and cloud
- **Early benchmarks:** 2–5× reduction in workflow setup time; measurable error reduction via automated validation and guardrails
- **Pipeline:** flagship chemistry workflows complete; design partner agreements in final stages across chemistry and a second vertical

See our demo here.

TEAM

- **Sanjay Vishwakarma (CEO):** Ex-IBM Quantum, PsiQuantum. Led enterprise platform initiatives spanning AI, distributed systems, and HPC. Brings the commercial and go-to-market motion for complex technical sales.
- **Trang Nguyen, PhD (CTO):** Built end-to-end molecular workflows at scale. Has lived the fragmentation problem firsthand, the rarest combination of domain depth and systems thinking.
- **Etinosa Osaro, PhD (CPO):** Product and systems engineering across complex technical platforms. Full-stack systems infrastructure engineer. HPC, Cloud, and AI deep expertise. Experience in founding companies.

WHY NOW

Three shifts make this the right moment to build Guavion: AI agents can now reason across multi-step tasks, but they still lack enterprise-grade execution infrastructure; compute is becoming more fragmented across cloud, HPC, and emerging architectures, increasing the need for a unifying orchestration layer; and regulated industries are raising the bar on reproducibility, traceability, and auditability. As AI moves closer to real operational work, the system that governs execution becomes just as important as the intelligence itself.

THE ASK

Guavion is raising a \$5 million seed round to ship production version 1, validate flagship verticals, and convert design partners into paying pilots. We are looking for investors who understand that the next major infrastructure category in AI is not just model access, but governed execution. Guavion is building the platform where work is designed, directed, executed, scaled, and continuously improved in one system.

Seeking AI + deep-tech investors to back the governed workflow operating system for the AI era.

Interested in knowing more? → www.guavion.com